Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211cs224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

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Campus Verse – Campus Event Portal

SDG 9: Industry, Innovation and Infrastructure

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INTRODUCTION

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- Campus event management is largely manual, causing missed opportunities and poor visibility for students.
- CampusVerse is a full-stack web portal centralising event discovery, registration, and admin for colleges.
- Students can browse events, register instantly, and manage their participation in real time.
- Admins manage events and view registrations through a dedicated, secure dashboard.
- Built with PHP 8.2, MySQL, Apache (XAMPP), custom CSS and JavaScript – no external frameworks.
- Security enforced via bcrypt, CSRF tokens, PDO prepared statements, and role-based session guards.

PROBLEM STATEMENT

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- Most colleges lack a unified digital platform for campus event management, forcing students and organizers to rely on notice boards, WhatsApp groups, and manual spreadsheets.
- Students miss events due to poor communication; admins face duplicate entries and zero attendance visibility.
- No real-time seat tracking leads to overbooking and no-shows during high-demand campus events.
- A secure, automated, role-based system is needed to manage campus events end-to-end in real time.

OBJECTIVE(S)

Objective 1:

- To design and develop a secure, role-based campus event portal enabling students to discover and register for events, and admins to manage events and track registrations in real time.

Objective 2:

- To implement robust security measures including bcrypt password hashing, CSRF protection, PDO prepared statements, session guards, and automatic slot enforcement to prevent overbooking.

PROPOSED SYSTEM

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- CampusVerse is a web-based campus event management portal with role-based access for students and admins.
- Students register for events instantly with real-time slot tracking; admins create, edit, and delete events from a dashboard.
- Replaces manual methods: no more notice boards, WhatsApp groups, or spreadsheets for event management.
- Secure dual-role authentication: separate login flows for students and administrators.
- Automated seat enforcement prevents overbooking without any manual intervention.
- Mobile-first responsive design accessible from any device – phone, tablet, or desktop.
- Admin analytics dashboard with live event statistics and full registration management.

PROPOSED SYSTEM ARCHITECTURE

System Flow:

- Input: Student/Admin login → Event registration form → Admin event creation form
- Processing: PHP 8.2 backend → PDO queries → CSRF/session validation → Bcrypt auth
- Database: MySQL (campus_event_portal) → Tables: users, admins, events, registrations
- Output: Student dashboard / Admin dashboard → Event listings → Registration confirmation

WORKFLOW

- **Step 1 – Homepage:** User visits portal, views events, clicks Sign In or Register.
- **Step 2 – Authentication:** Student or Admin selects role, submits credentials, PHP validates, session created.
- **Step 3 – Student Flow:** Dashboard → Browse Events → Register → Slot verified → DB insert → Confirmation.
- **Step 4 – Admin Flow:** Dashboard → Add/Edit/Delete Events → View Registrations → Filter by event/dept.
- **Step 5 – Logout:** Session destroyed, cookie cleared, redirect to homepage.

TECHNOLOGIES USED

- **Frontend:** HTML5, CSS3 (Custom – no frameworks), Vanilla JavaScript, Google Fonts (Plus Jakarta Sans), Responsive Design
- **Backend:** PHP 8.2 (Pure PHP – no frameworks), PDO for database access, Session-based authentication, bcrypt password hashing
- **Database:** MySQL 8 (MariaDB via XAMPP), 4 tables: users, admins, events, registrations, utf8mb4 charset
- **Tools:** XAMPP (Apache 2.4 + MySQL), phpMyAdmin, VS Code, Git, Chrome DevTools
- **Hardware/Software Requirements:** Windows 10/11 OS, 4GB RAM minimum, XAMPP v8.2, Any modern browser (Chrome/Firefox/Edge)

MODULE DESCRIPTION

CampusVerse has 4 core modules:

Module 1: Authentication & User Management
Handles student and admin registration, dual-role login (bcrypt + CSRF + session), password strength validation, and secure logout with session destruction.

Module 2: Event Management (Admin)
Allows admins to create, edit, and delete events with full details (title, category, date, location, price, max seats, organizer). Real-time registration count per event.

Module 3: Event Registration (Student)
Students browse upcoming events, register with one click, and cancel registrations. Automatic slot enforcement prevents overbooking via DB constraint.

Module 4: Admin Dashboard & Reports
Admin dashboard shows live stats (total events, students, registrations, upcoming count). View all registrations with search and filter by event or department.

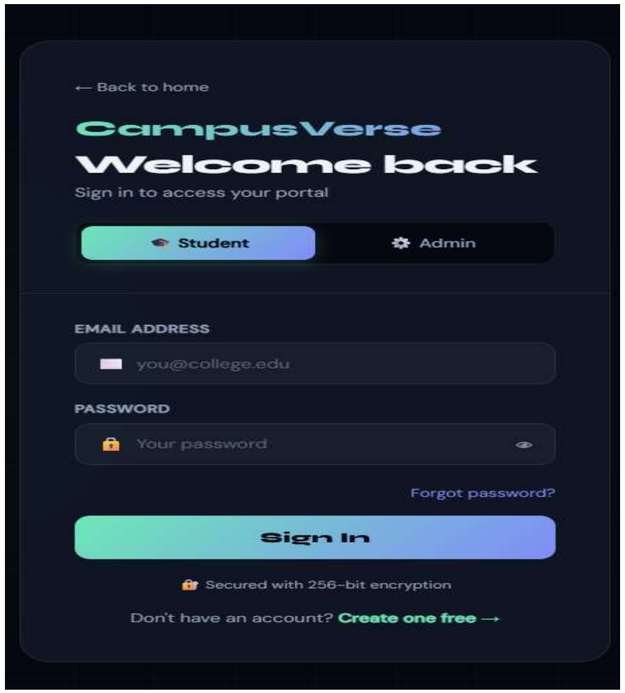
IMPLEMENTATION SCREENSHOTS

Home Page:

- The CampusVerse landing page features a dark space-themed hero section with animated gradient orbs, a scrolling event ticker, live counters for total events/students/registrations, and upcoming event cards.

Login Page:

- The login page features a glassmorphism card design with a dual-role toggle (Student / Admin), email/password fields with show/hide toggle, CSRF token protection, and animated card entrance.



IMPLEMENTATION SCREENSHOTS

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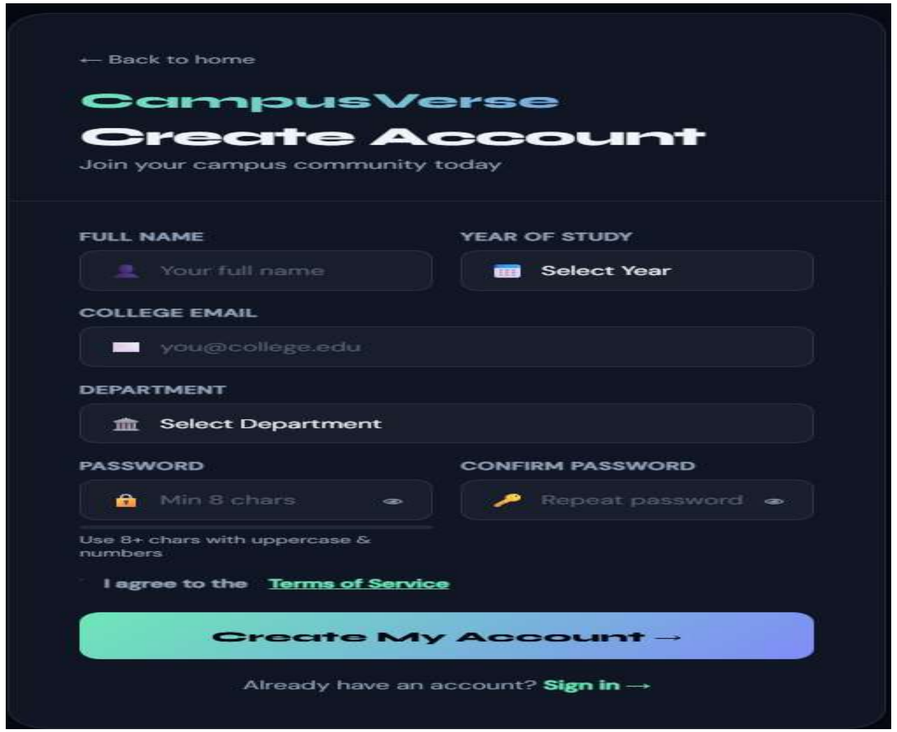
Student Registration Page:

The registration form includes full name, email, department dropdown (9 options), year of study, password with live strength meter (Weak/Fair/Good/Strong), confirm password match check, and terms checkbox.



IMPLEMENTATION SCREENSHOTS

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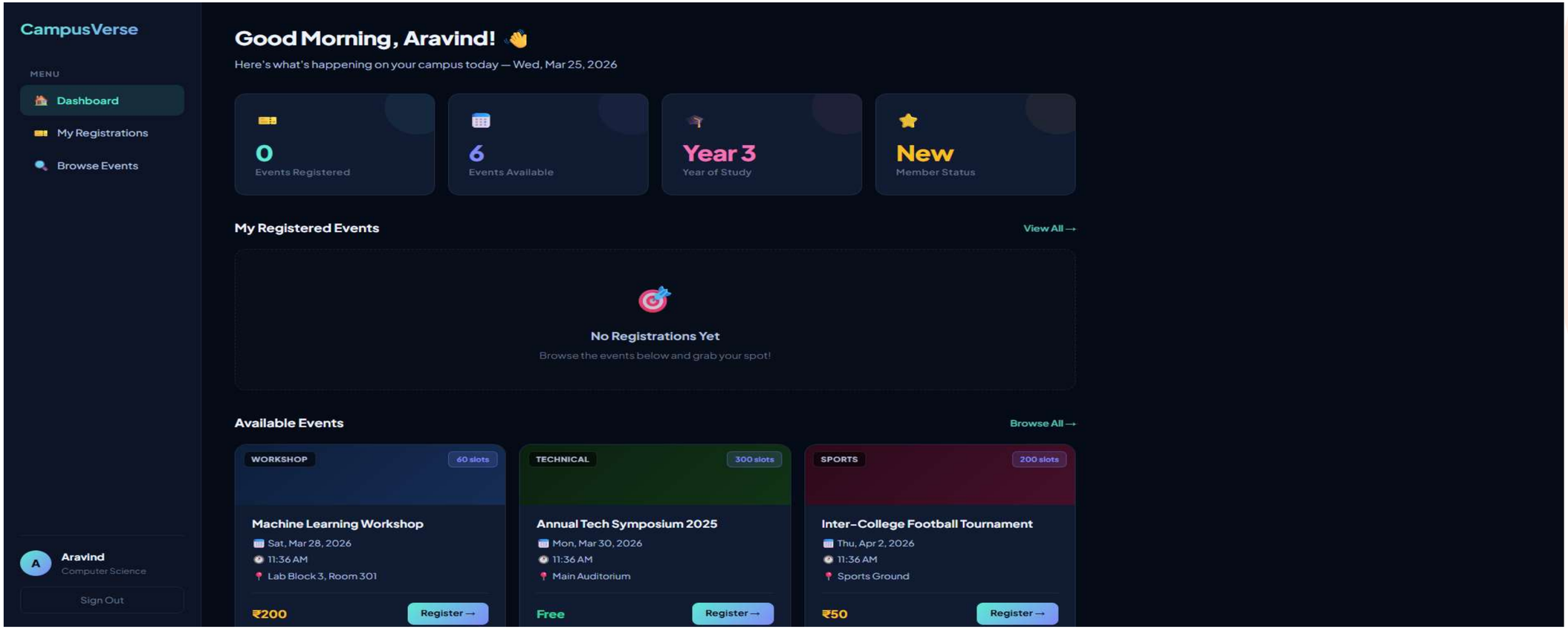
Student Dashboard:

- The student dashboard shows a personalised greeting, 4 stat cards (events registered, available, year, member status), registered event cards with cancel option, and available events with register buttons.



IMPLEMENTATION SCREENSHOTS

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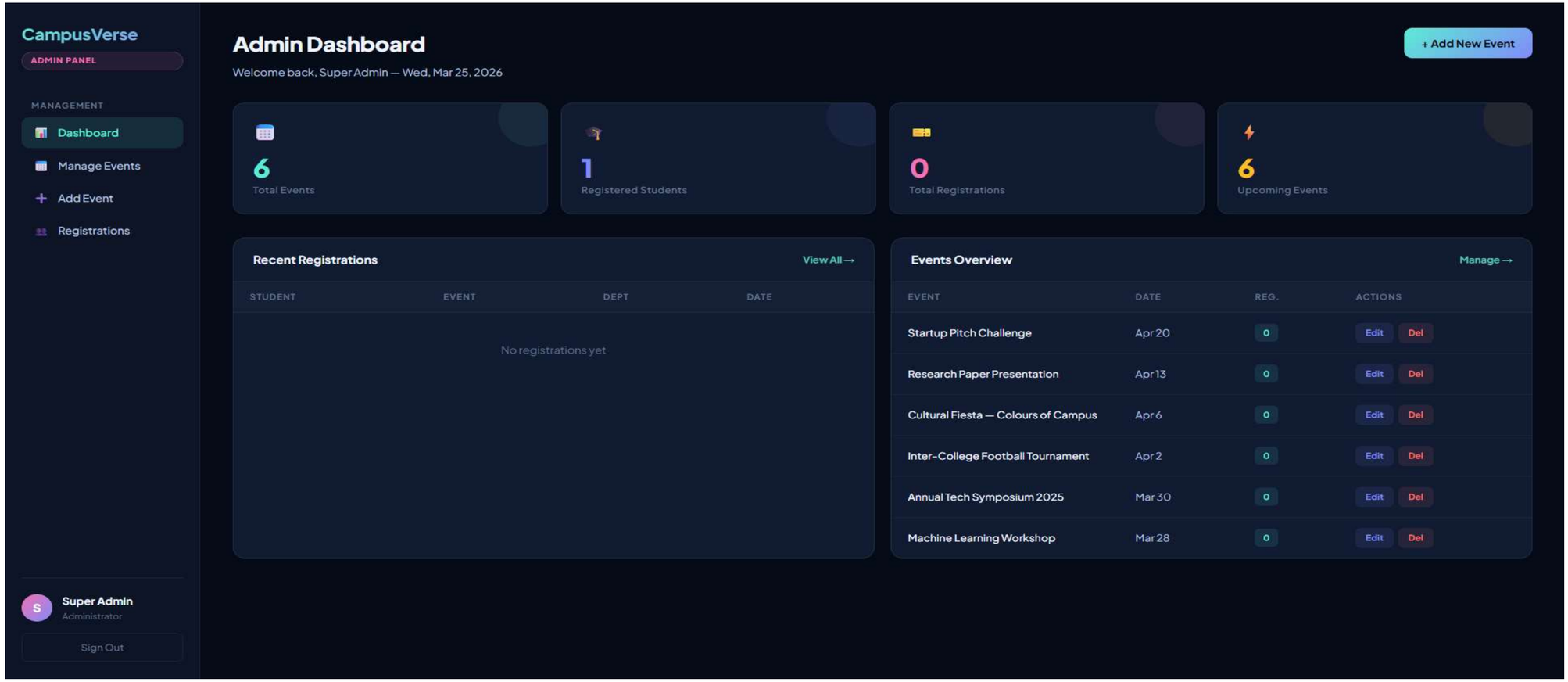
Admin Dashboard

- Shows live stats: Total Events, Students, Registrations, and Upcoming Events in 4 cards.
- Recent registrations table displays student name, event, department, and date instantly.
- Events overview table provides Edit and Delete buttons for quick event control.



ADVANTAGES & LIMITATIONS

ADVANTAGES:

- Fully automated end-to-end event management eliminating manual processes like notice boards and spreadsheets.
- Dual-role authentication with industry-standard security: bcrypt hashing, CSRF tokens, PDO prepared statements.
- Real-time seat availability enforcement prevents overbooking with zero manual intervention.
- Mobile-first responsive UI accessible on any device without installing a separate app.
- Built with pure PHP and MySQL – no external frameworks, making it lightweight, portable, and easy to deploy.

LIMITATIONS:

- No email notification system – students do not receive automated confirmation emails upon registration.
- No payment gateway integration – paid events require manual fee collection outside the portal.
- Currently single-institution – no multi-college or multi-campus support in the current version.
- No QR code-based attendance tracking or digital pass generation for event entry.

Thank you